

(TR-104) PYTHON WITH AI/ML

Training Day 42 Report

Date: 10 March 2026

Topic Covered: RAG API Optimization, Redis Caching & Security Enhancements

The forty-second day of training focused on improving the RAG-based AI Assistant API by implementing caching, rate limiting, security mechanisms, and integrating production-ready FastAPI features.

- Continued integrating the fine-tuned generative model into the existing RAG pipeline.
- Implemented Redis-based caching for storing responses to repeated questions.
- Reduced unnecessary pipeline execution by serving cached responses for repeated queries.
- Created a custom cache key builder using normalized questions and SHA256 hashing for consistent cache management.
- Integrated SlowAPI rate limiting to prevent API abuse and restrict excessive requests.
- Configured API rate limit to allow only 10 requests per minute per client IP.
- Added a global exception handler for rate limit violations with proper HTTP 429 responses.
- Implemented XSS protection by sanitizing user inputs using the bleach library.
- Configured CORS middleware for secure cross-origin API communication.
- Set up FastAPI lifespan events for automatic Redis connection initialization and cleanup during startup and shutdown.
- Strengthened understanding of production-level API optimization and secure AI application design.
- Learned how caching improves response time and reduces repeated computation in RAG systems.

- Explored security mechanisms including rate limiting, sanitization, exception handling, and middleware integration in FastAPI applications.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/projects/blob/main/app.py>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/supercharge-your-ai-hands-on-generative-fine-tuning-04ddb530be9d>